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Bihar Engineering University, Patna

B.Tech 6th Semester Exam., 2026

Code: 100601		***Time: 03 Hours***
Semester: VI	***Subject: Biology for Engineers***	***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) Which of the given consists of only 'unicellular prokaryotic organisms'?
(i) Fungi (iii) Monera
(ii) Protista (iv) Plantae

(b) Where is the genetic information found?
(i) Centriole (iii) Golgi Apparatus
(ii) Nucleus (iv) Ribosomes

(c) Which of the following is a chemical basis of heredity in the cell?
(i) Mitochondria (iii) RNA
(ii) DNA (iv) Ribosomal RNA

(d) Which major factor differentiates DNA from RNA?
(i) Guanine (iii) Thymine
(ii) Cytosine (iv) Uracil

(e) Which of the following is not an exclusive characteristic of living organisms?
(i) Movement (iii) Metabolism
(ii) Reproduction (iv) Consciousness

(f) What is the nature of charges that amino acid side chains carry?
(i) Positive (iii) Neutral
(ii) Negative (iv) A and B

(g) Branch of science dealing with classification of organisms is called as _____
(i) Bioautonomy (iii) Nomenclature
(ii) Taxonomy (iv) Specialization

(h) Who is known as the "Father of Genetics"?
(i) Mendel (iii) Watson
(ii) Bateson (iv) Morgan

(i) The polypeptide chain of the amino acid sequence of a protein is called as _____
(i) Quaternary structure (iii) Tertiary structure
(ii) Secondary structure (iv) Primary structure

(j) The basic building blocks of proteins are
(i) Sugars (iii) Amino acids
(ii) Fatty acids (iv) Nucleotides

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Q.2 Solve both questions : **[7 x 2 = 14]**

(a) Explain the Laws of Thermodynamics in biological systems.

(b) Compare Eye with Camera.

Q.3 Solve both questions : **[7 x 2 = 14]**

(a) Compare Prokaryotic and Eukaryotic cell with suitable diagram.

(b) What are Proteins? Write any four functions of proteins.

Q.4 Solve both questions : **[7 x 2 = 14]**

(a) What are enzymes and explain its role in plants?

(b) How autotrophs utilize carbon and energy?

Q.5 Solve both questions : **[7 x 2 = 14]**

(a) What is Mitotic Cell division? Explain Mitosis with neat diagram.

(b) Give brief account on Dominant Epistasis With suitable example.

Q.6 Solve both questions : **[7 x 2 = 14]**

(a) Compare Bird flying and aircraft.

(b) Write short notes on fundamental differences between science and engineering.

Q.7 Solve both questions : **[7 x 2 = 14]**

(a) What is the concept of single celled organisms.

(b) Describe Glycolysis in detail with suitable diagram.

Q.8 Solve both questions : **[7 x 2 = 14]**

(a) Write a note on sterilization and various techniques used.

(b) Define Biology. Explain Terrestrial Habitat.

Q.9 Write short notes on any two of the following: **[7 x 2 = 14]**

(a) Growth Kinetics

(b) Photosynthesis

(c) Microscopic techniques

(d) Brownian Motion

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